

Visual Question Answering on VizWiz with a Self-Designed Cross-Modal Attention

DL Basic 2026 Spring — Final Assignment (Competition)

1. Overview

We build a Visual Question Answering (VQA) model from scratch for the VizWiz 2023 split (19,873 train / 4,969 test). Given an image and a question, the model predicts an answer string; performance is measured with the standard VQA accuracy $\text{Acc}(\text{ans}) = \frac{\text{mean}_i \min(\#\text{annotators}_{\{j \neq i\}} \text{ that said ans } / 3, 1)}{1}$.

Our model is a modular, config-driven architecture: a swappable **image encoder** and **text encoder** produce token/region sequences that are combined by a swappable **fusion** module and classified over a closed answer vocabulary. Pretrained backbones (ResNet, BERT) are used *only as components* and fine-tuned by our own training loop — no end-to-end VQA model is used. The contribution highlighted here is a **self-designed cross-modal attention fusion**, evaluated against a concat baseline in an ablation study.

2. Method

2.1 Encoders (unified (features, mask) interface)

- **Image** (swappable): scratch ResNet18 (baseline), ImageNet-pretrained ResNet50, ViT-B/16, or a CLIP ViT-B/16 vision encoder. For CNNs we keep the spatial feature map before global pooling; for ViT/CLIP we keep the patch tokens — both give region tokens $V \in \mathbb{R}^{B \times N \times d_v}$. Normalization matches the backbone (ImageNet stats, or CLIP stats for the CLIP encoder).
- **Text**: the baseline encodes the question as a multi-hot vocabulary vector projected by a linear layer (one token, order-insensitive). The improved model uses fine-tuned **BERT**, giving contextual token embeddings $Q \in \mathbb{R}^{B \times L \times d_q}$ with an attention mask.

2.2 Cross-modal attention fusion (centerpiece)

Both modalities are projected to a shared width H : $V' = V W_v$, $Q' = Q W_q$. The question tokens **attend over the image regions** with multi-head attention

```
A = MultiHeadAttention(query = Q', key = V', value = V')      # (B, L, H)
Q_ctx = LayerNorm(Q' + A)                                     # residual keeps the question
signal
```

We then pool a masked mean of the attended question tokens and of the image regions and fuse them:

```
z = MLP( [ mean_mask(Q_ctx) ; mean(V') ] )                  # (B, H)
logits = W_cls z
```

Intuition: instead of squashing each modality to one vector and concatenating (which discards *where* in the image the question refers to), each question token pulls the image regions it needs, so the fused representation is grounded in question-relevant image content.

2.3 Training

Closed-vocabulary classifier over the full set of normalized training answers (no catch-all `<unk>` class — a catch-all funnels the long answer tail into one dominant, always-wrong class and inflates index-based validation). Loss is cross-entropy on the most-frequent answer (hard label) or the VQA-score soft-label

distribution over the 10 annotators (soft label). Following the course fine-tuning practice: AdamW with **differential learning rates** (pretrained encoders at $2e-5$, the freshly-initialized fusion + head at $5e-4$), cosine schedule with warmup, gradient clipping (max-norm 1.0), and AMP. Augmentation is RandomResizedCrop + ColorJitter only — we drop horizontal flip, which can invert the meaning of spatially/text-dependent questions. A 10% held-out slice of train gives the VQA-accuracy validation below; the placeholder class is masked at evaluation and inference to mirror the string-matched test.

3. Experiments

Dataset, metric, and closed-vocabulary setup as above. We ablate three axes: image encoder (scratch vs pretrained), fusion (concat vs cross-attention), and label (hard vs soft). Validation VQA accuracy is on our held-out train slice; the test column is the Omnicampus score.

Ablation table

#	Image encoder	Text encoder	Fusion	Label	Extras	Val VQA acc
0	ResNet18 (scratch)	one-hot	concat	hard	—	0.499 (official baseline, test)
1	ResNet50 (pretrained)	one-hot	concat	hard	—	0.5005
2	ResNet50 (pretrained)	BERT	cross-attention	soft	—	0.5391
3	ViT-B/16 (pretrained)	BERT	cross-attention	soft	—	0.5534 (test 0.55092)
4	CLIP ViT-B/16	BERT	cross-attention	soft	diff-LR	0.5232

Progression: a pretrained image encoder alone barely helps (0→1, +0.1). Replacing one-hot with BERT and concat with cross-attention (+ soft labels) is the largest jump (1→2, **+3.9 points**); a stronger ViT backbone adds a further +1.4 (2→3). Row 3 is our submitted model: held-out validation 0.5534, **Omnicampus test 0.55092** — well above the 0.499 completion line and close to the ~0.60 target. A CLIP vision encoder (row 4), despite being vision-language aligned, under-performed ViT at equal epochs — its validation was still rising at epoch 5 (under-converged) rather than clearly better, so ImageNet-pretrained ViT remained our best backbone in the available budget. (Row 0 is the official-baseline Omnicampus test score; rows 1–4 are our held-out validation except where a test score is given.)

Key finding

Upgrading only the image encoder (row 0→1) barely moved the honest validation metric (0.499 → 0.5005): with a one-hot text encoder the model defaults to the dominant "unanswerable" answer. This localizes the bottleneck to the **text representation and fusion**. Replacing one-hot with fine-tuned BERT and concat with the cross-attention fusion (plus soft labels) then delivers the main gain (row 2, +3.9 points), and a stronger ViT backbone adds a further +1.4 (row 3, best 0.5534) — confirming the diagnosis.

4. Discussion

The image-encoder upgrade alone (row 0→1) is nearly a no-op on the honest metric, while adding BERT + cross-attention + soft labels (row 2) yields +3.9 points — the text representation and cross-modal grounding, not the visual backbone, are where VizWiz accuracy is won. The model still over-predicts the majority "unanswerable" class (~74% of predictions; a known VizWiz characteristic and a class-imbalance artifact — outputting the majority answer minimizes loss), which bounds accuracy. Two levers we identified for going further: (i) re-weighting the loss by inverse answer frequency to stop the model from defaulting to the majority class, and (ii) an open-vocabulary / generative answer head, since the closed-vocabulary classifier structurally cannot emit answers absent from the training set.

5. Conclusion

A modular from-scratch VQA model with a self-designed cross-modal attention fusion. The ablation isolates where accuracy comes from — text representation + cross-modal fusion, not the visual backbone alone. The submitted ViT-B/16 + BERT + cross-attention + soft-label model reaches **0.55092 VQA accuracy on the Omnicampus test set**, clearing the 0.499 completion line by ~5 points. Pretrained encoders are used only as fine-tuned components inside our own model and training loop. Remaining headroom toward ~0.60 lies in curbing the majority-class bias (loss re-weighting) and an open-vocabulary generative head — our identified next steps.